



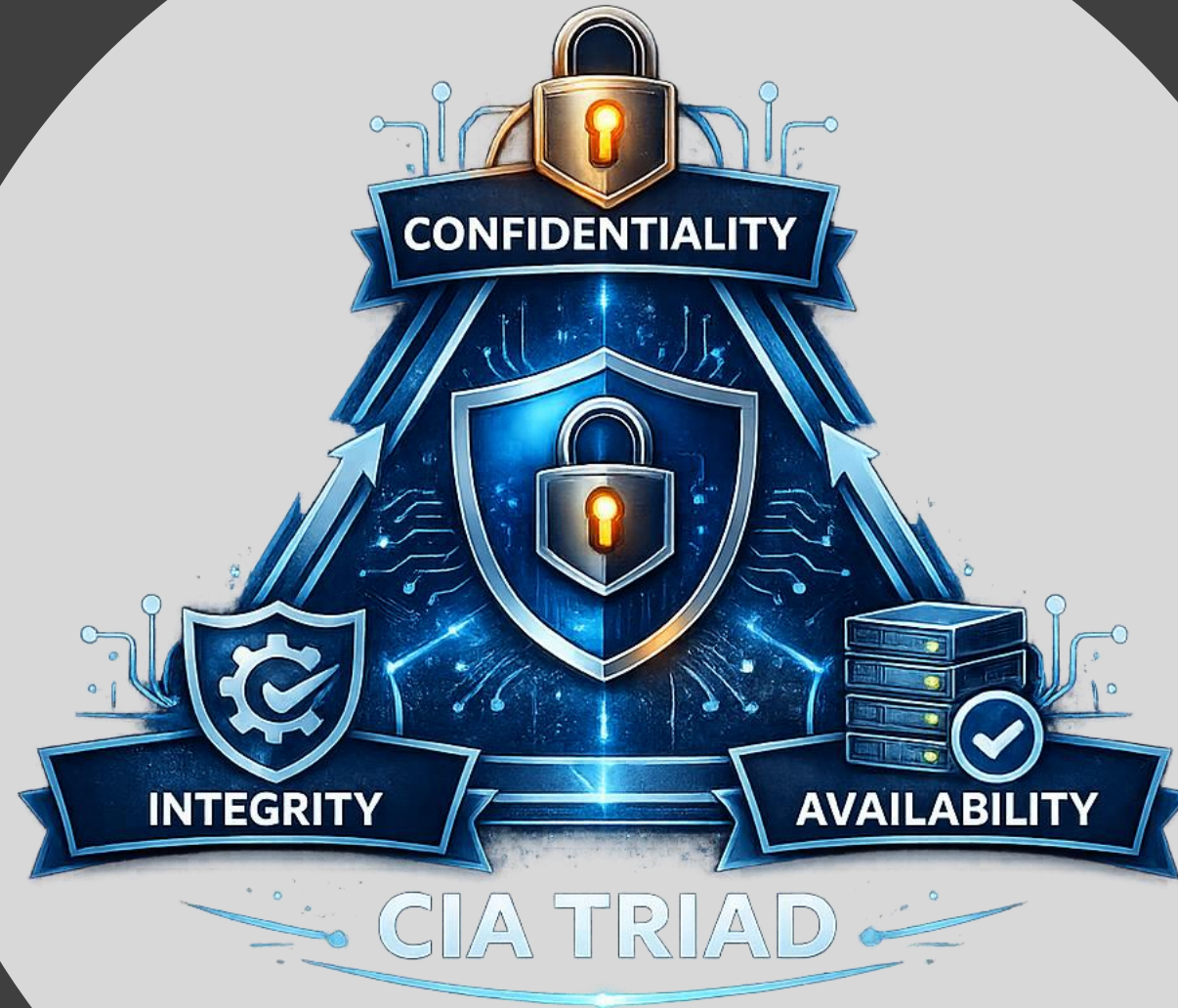
Basic Concepts in Security

CNS 2.1 – 2.2

NSE 1.1 – 1.4

Goals of Network Security

– CIA Triad

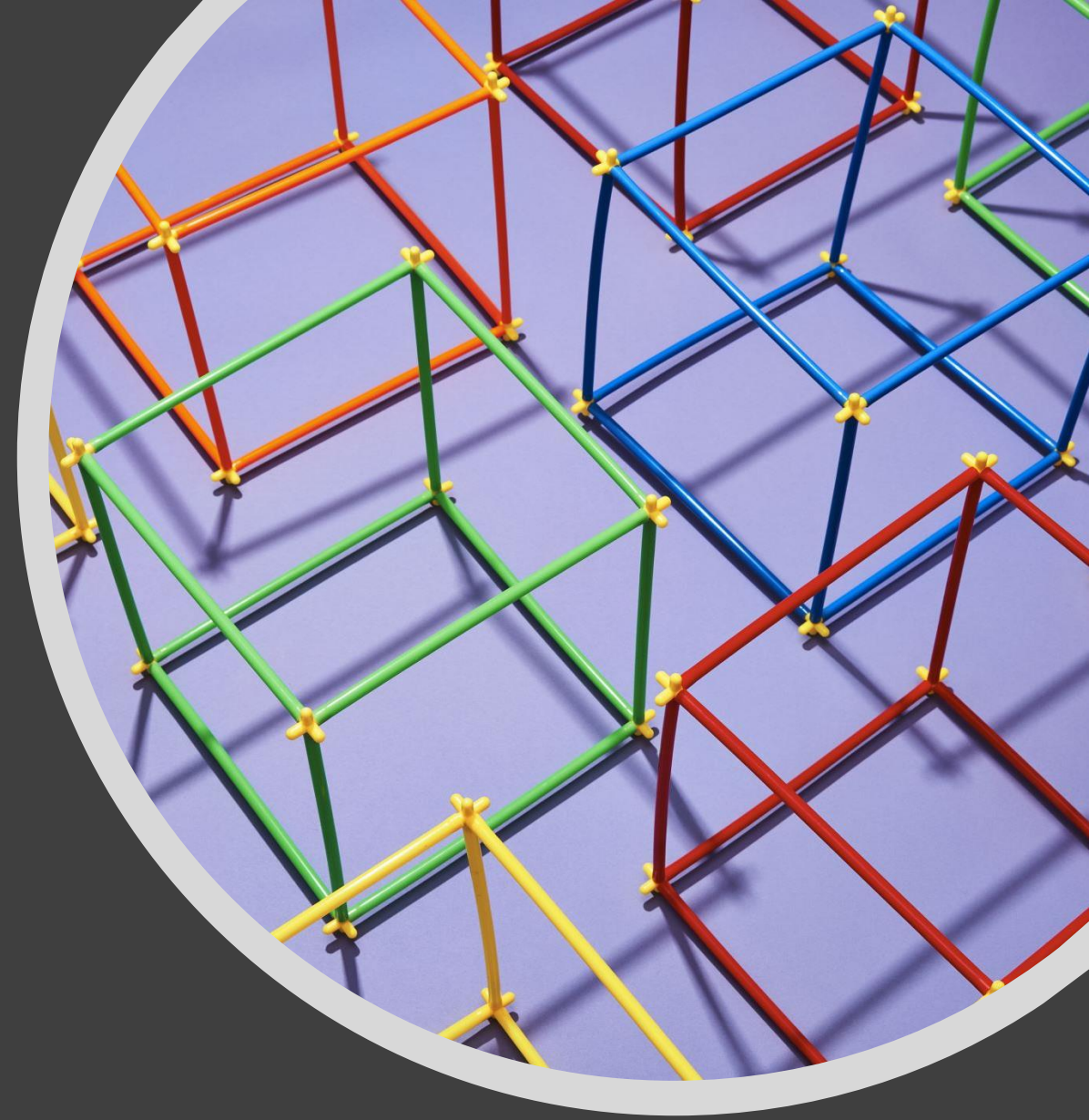


Security Architecture for OSI

- **ITU-T Recommendation X.800**, Security Architecture for OSI defines systematic way to:
 - Define the requirements for security
 - Characterizing the approaches to satisfying those requirements
- *ITU-T – International Telecommunication Union Telecommunication Standardization Sector*
- *OSI – Open Systems Interconnection – an effort started in 1977 to standardize computer networking*

OSI Security Architecture

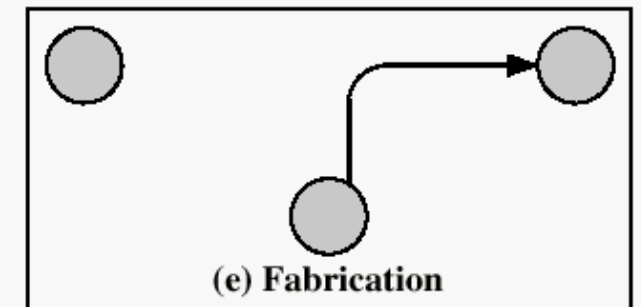
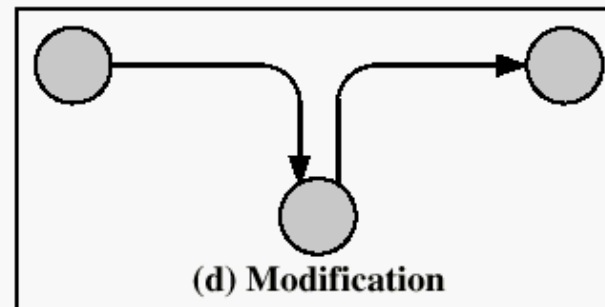
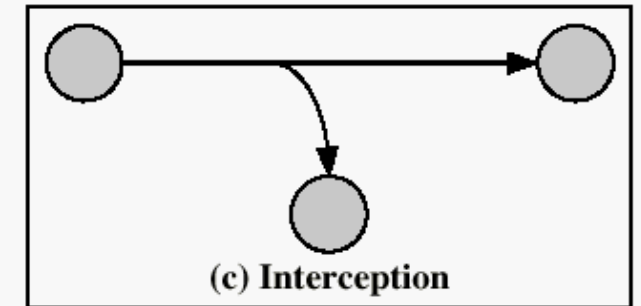
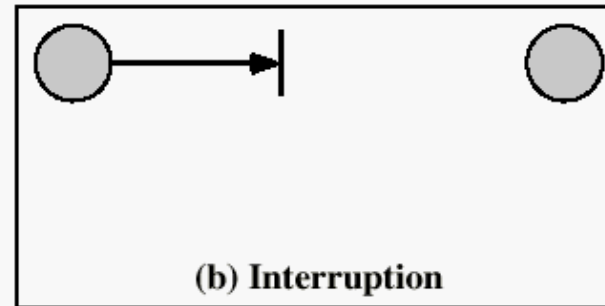
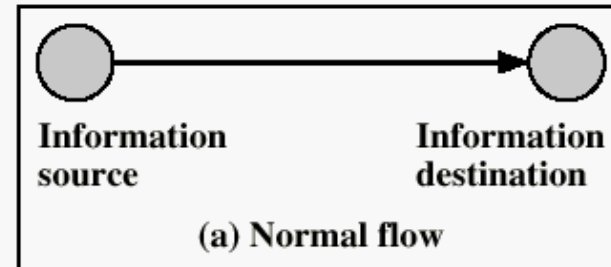
- The following concepts are used:
- **Security attack:** Any actions that compromises the security of information owned by an organization (or a person).
- **Security mechanism:** a mechanism that is designed to detect, prevent, or recover from a security attack.
- **Security service:** a service that enhances the security of the data processing systems and the information transfers of an organization. The services make use of one or more security mechanisms to provide the service



Security attacks

- **Passive attack:** aims to learn or make use of information from the system but does not affect system resources. (prevent)(quantum)
- **Active attack:** attempts to alter system resources or affect their operation. (detect)

Security Attacks



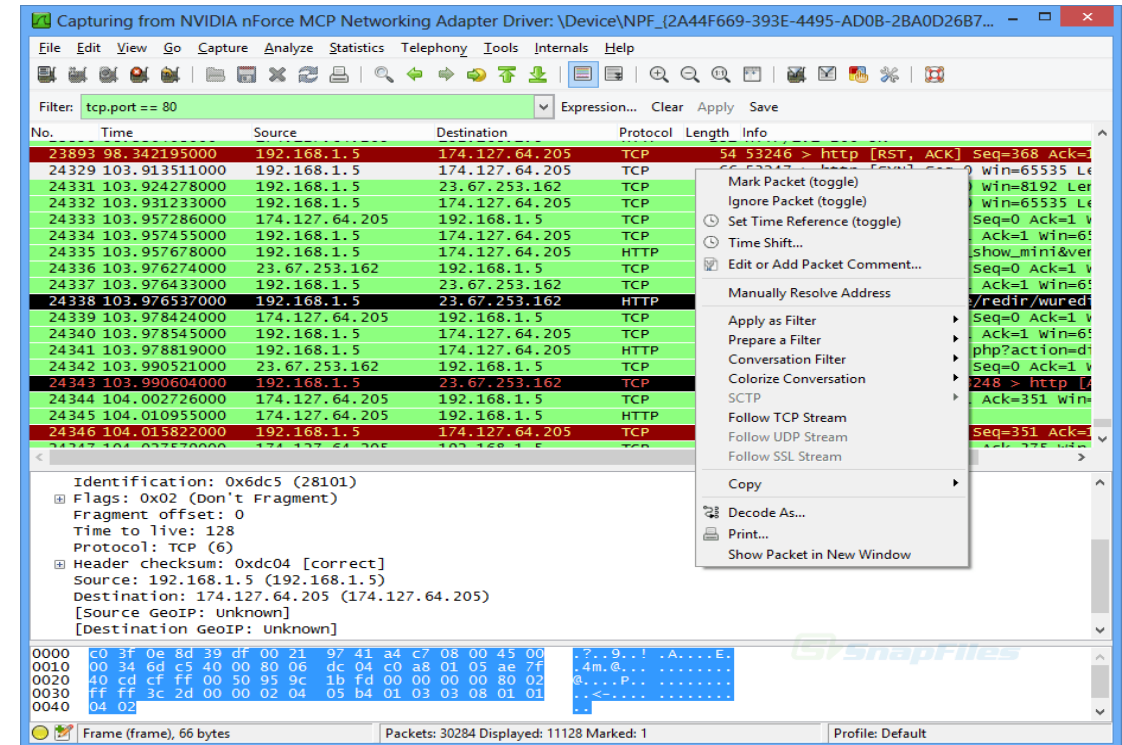
Passive Attacks

- The Interception of a message without modification

- It may be read or logged

- **Traffic analysis** is useful in many cases

HarvestNowDecryptLater, Quantum comp.



```
root@kali:~# nmap -sn 192.168.1.100

Starting Nmap 7.60 ( https://nmap.org ) at 2018-01-09 03:00 EST
Nmap scan report for 192.168.1.100
Host is up (0.00016s latency).
MAC Address: FC:AA:14:6A:9A:A2 (Giga-byte Technology)
Nmap done: 1 IP address (1 host up) scanned in 0.12 seconds
root@kali:~#
```

Security Attacks

- **Interruption:** This is an attack on availability
- **Interception:** This is an attack on confidentiality
- **Modification:** This is an attack on integrity
- **Fabrication:** This is an attack on authenticity

Security services/security attributes

- Security service is a service which ensures adequate security (a particular security attribute) of the systems or of data transfers
- X.800 Recommendation divides security services into 6 categories:
 - Authentication
 - Access control
 - Data confidentiality
 - Data integrity
 - Non-repudiation
 - Availability

Authentication

The authentication service is concerned with assuring that communication is authentic:

The recipient of the message should be sure that the message came from the source that it claims to be

All communicating parties should be sure that the connection is not interfered with by unauthorized party.

Example: consider a person, using online banking service. Both the user and the bank should be assured in identities of each other

Access control

This
service
controls

who can have access to a resource;
under what conditions access can occur;
what those accessing are allowing to do.



Example:

an online banking a user may be allowed
to see his balance but not allowed to
make any transactions for a joint
account.

Data confidentiality

- The protection of data from unauthorized disclosure
 - Connection confidentiality
 - Connectionless confidentiality
 - Selective field confidentiality
 - Traffic-Flow confidentiality

Data Integrity

- The assurance that data received are exactly as sent by an authorized entity, i.e. contain
 - no modification
 - no insertion
 - no deletion
 - no replay

Non- repudiation

- Protection against denial by one of the entities involved in a communication of having participated in the communication.
- Nonrepudiation can be related to
 - **Origin:** proof that the message was sent by the specified party
 - **Destination:** proof that the message was received by the specified party

Example: Imagine a user of online banking who has made a transaction but later denied that. How can the bank can protect itself in a such situation?

Availability service

- Protects a system to ensure its availability
- Particularly, it addresses denial-of-service attacks
- Depends on other security services: access control, authentication, etc

Attacks and Security services (X.800)

		Attack					
		Release of message contents	Traffic analysis	Masquerade	Replay	Modification of messages	Denial of service
Service	Peer Entity Authentication			Y			
	Data origin authentication			Y			
	Access control			Y			
	Confidentiality	Y					
	Traffic flow confidentiality		Y				
	Data integrity				Y	Y	
	Non-repudiation						
	Availability						Y

Security mechanisms

- Security mechanisms are used to implement security services. They include (X.800):
 - Encipherment or Encryption
 - Digital signature
 - Access Control mechanisms
 - Data Integrity mechanisms (e.g., Hash)
 - Authentication Exchange
 - Traffic Padding
 - Routing Control
 - Notarisation